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Bulk cell transcriptomic data analysis of endometrial cancer cell

Data source: https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi

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Scope: Self | Format: HTML | Amount: Quick | GEO accession: GSE247791 | GO

Series GSE247791 — Query DataSets for GSE247791

Status	Public on Jan 21, 2025
Title	Cell cycle checkpoint proficiency of endometrial cancer cells [bulk RNA-Seq]
Organism	Homo sapiens
Experiment type	Expression profiling by high throughput sequencing
Summary	Endometrial cancers (EC) are the most common gynecologic malignancy in the US. Most ECs harbor limited targetable somatic alterations, and are often grouped by histology (endometrioid, serous, or clear cell), mismatch repair, or TP53 status, none of which perfectly predict therapeutic response. A better mechanistic understanding of the key functional defects in ECs and more therapies with which to engage those targets in advanced stage EC are needed. Here we utilize functional, transcriptomic and genomic assays on a panel of EC cell lines and patient-derived organoids across EC histologic and genomic subtypes to characterize the TP53 and RB1 cell cycle regulatory proficiency and therapeutic vulnerabilities in this disease. We were surprised to find that TP53 genomic and functional status has little predictive capacity for EC therapeutic response. Rather, RB regulatory status correlated better with response to G1/S targeted therapies.

Download family	Format
SOFT formatted family file(s)	SOFT
MINiML formatted family file(s)	MINiML
Series Matrix File(s)	TXT

Supplementary file	Size	Download	File type/resource
GSE12453_RAW.tar	468.3 Mb	(http)(custom)	TAR (of CEL)

Processed data included within Sample table

###BULK CELL TRANSCRIPTOMIC###
#Najneen Rejwana
#count data: https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE247791

```
#metadata: https://www.ncbi.nlm.nih.gov/Traces/study/?acc=PRJNA1040329&o=acc_s%3Aa

setwd("G:/Rstudio/TRANSCRIPTOMICS/bulk_trans/GSE247791")
getwd()


library(DESeq2)
library(ggplot2)
library(pheatmap)
library(EnhancedVolcano)
library(ggrepel)

count_data <- read.csv("count_data_GSE247791_EDIT.csv", row.names = 1)
meta_data <- read.csv("meta_data_GSE247791_EDIT.csv", row.names = 1)
View(count_data)
View(meta_data)

str(count_data)
str(metadata)

# Check if sample names in count matrix match metadata row names
all(colnames(count_data) == rownames(meta_data))
#[1] FALSE


setdiff(colnames(count_data), rownames(meta_data))  # Samples in count_matrix but not in
metadata
#[1] "ARK1_Abema_A"   "ARK1_Abema_B"   "ARK1_DMSO_A"    "ARK1_DMSO_B"
"HEC1B_Abema_A"
#[6] "HEC1B_Abema_B"  "HEC1B_DMSO_A"   "HEC1B_DMSO_B"   "EMCA.A_DMSO_B"
"EMCA.A_DMSO_A"
#[11] "EMCA.A_Abema_B" "EMCA.A_Abema_A" "CS.A_DMSO_B"    "CS.A_DMSO_A"
"CS.A_Abema_B"
#[16] "CS.A_Abema_A"   "UPSC.A_DMSO_B"  "UPSC.A_DMSO_A"  "UPSC.A_Abema_B"
"UPSC.A_Abema_A"
#[21] "UPSC.A_DMSO_1"  "UPSC.A_DMSO_2"  "CS.A_DMSO_1"    "CS.A_DMSO_2"
"EMCA.A_DMSO_1"
#[26] "EMCA.A_DMSO_2"  "CS.B_DMSO_1"    "CS.B_DMSO_2"    "AN3_DMSO1"
"AN3_DMSO2"
```

```
#[31] "ARK1_DMSO1"     "ARK1_DMSO2"     "HEC_DMSO1"      "HEC_DMSO2"
"KLE_DMSO1"
#[36] "KLE_DMSO2"
setdiff(rownames(meta_data), colnames(count_data))  # Samples in metadata but not in
count_matrix
#[1] "SRR26830900" "SRR26830901" "SRR26830904" "SRR26830905" "SRR26830908"
"SRR26830909"
#[7] "SRR26830912" "SRR26830913" "SRR26830916" "SRR26830917" "SRR26830920"
"SRR26830921"
#[13] "SRR26830928" "SRR26830929" "SRR26830932" "SRR26830933" "SRR26830936"
"SRR26830937"
#[19] "SRR26830938" "SRR26830939" "SRR26830940" "SRR26830941" "SRR26830942"
"SRR26830943"
#[25] "SRR26830948" "SRR26830949" "SRR26830950" "SRR26830951" "SRR26830952"
"SRR26830953"
#[31] "SRR26830954" "SRR26830955" "SRR26830956" "SRR26830957" "SRR26830958"
"SRR26830959"

count_data_edit <- read.csv("count_data_GSE247791_EDIT.csv", row.names = 1)
meta_data_edit <- read.csv("meta_data_GSE247791_EDIT.csv", row.names = 1)
View(count_data_edit)
View(meta_data_edit)

str(count_data_edit)
str(meta_data_edit)


ncol(count_data_edit)  # Number of samples in count matrix
#[1] 36
nrow(meta_data_edit)   # Number of samples in metadata
#[1] 36
all(colnames(count_data_edit) == rownames(meta_data_edit))
#[1] FALSE


print(colnames(count_data_edit))
# [1] "SRR26830959" "SRR26830958" "SRR26830957" "SRR26830956" "SRR26830955"
"SRR26830954"
#[7] "SRR26830953" "SRR26830952" "SRR26830951" "SRR26830950" "SRR26830949"
"SRR26830948"
```

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```
#[13] "SRR26830943" "SRR26830942" "SRR26830941" "SRR26830940" "SRR26830939"
"SRR26830938"
#[19] "SRR26830937" "SRR26830936" "SRR26830933" "SRR26830932" "SRR26830929"
"SRR26830928"
#[25] "SRR26830921" "SRR26830920" "SRR26830917" "SRR26830916" "SRR26830913"
"SRR26830912"
#[31] "SRR26830909" "SRR26830908" "SRR26830905" "SRR26830904" "SRR26830901"
"SRR26830900"
print(rownames(meta_data_edit))
#[1] "SRR26830900" "SRR26830901" "SRR26830904" "SRR26830905" "SRR26830908"
"SRR26830909"
#[7] "SRR26830912" "SRR26830913" "SRR26830916" "SRR26830917" "SRR26830920"
"SRR26830921"
#[13] "SRR26830928" "SRR26830929" "SRR26830932" "SRR26830933" "SRR26830936"
"SRR26830937"
#[19] "SRR26830938" "SRR26830939" "SRR26830940" "SRR26830941" "SRR26830942"
"SRR26830943"
#[25] "SRR26830948" "SRR26830949" "SRR26830950" "SRR26830951" "SRR26830952"
"SRR26830953"
#[31] "SRR26830954" "SRR26830955" "SRR26830956" "SRR26830957" "SRR26830958"
"SRR26830959"
```

###It seems that the order of the sample names between count_data_edit and meta_data_edit is reversed. The sample names are present in both, but they are in a different order.

```
#Align the Order:
count_data_edit <- count_data_edit[, order(colnames(count_data_edit))]
meta_data_edit <- meta_data_edit[order(rownames(meta_data_edit)), , drop=FALSE]

all(colnames(count_data_edit) == rownames(meta_data_edit))
#[1] TRUE
```

```
####DSEQ2 analysis-----------------------------------------------------------------
dds <- DESeqDataSetFromMatrix(countData = count_data_edit,
                              colData = meta_data_edit,
                              design = ~ treatment)
```

```
#Warning message:
#In DESeqDataSet(se, design = design, ignoreRank) :
# some variables in design formula are characters, converting to factor


#Check conversion
str(meta_data_edit$treatment)  #before conversion
#chr [1:36] "DMSO" "DMSO" "DMSO" "DMSO" "DMSO" "DMSO" "DMSO" "DMSO"
"DMSO" "DMSO" "DMSO" ...

##After conversion
# Convert to factor if not already done
meta_data_edit$treatment <- factor(meta_data_edit$treatment)

# Show structure again
str(meta_data_edit$treatment)
#Factor w/ 2 levels "Abemaciclib",..: 2 2 2 2 2 2 2 2 2 2 ...
#It looks like the treatment column has been successfully converted into a factor with two levels:
"Abemaciclib" and "DMSO". The numbers (e.g., 2 2 2 2 2 ...) represent the factor levels (where
"Abemaciclib" is 1 and "DMSO" is 2 in the factor encoding).


table(meta_data_edit$treatment)   #Check how many samples fall under each treatment group,
#Abemaciclib       DMSO
#10                26




############# Run DESeq2 analysis#####################################
dds <- DESeqDataSetFromMatrix(countData = count_data_edit,
                              colData = meta_data_edit,
                              design = ~ treatment)

# Pre-filter low count genes (optional but recommended)
dds <- dds[rowSums(counts(dds)) > 1, ]

# Run the DESeq function
dds <- DESeq(dds)
```

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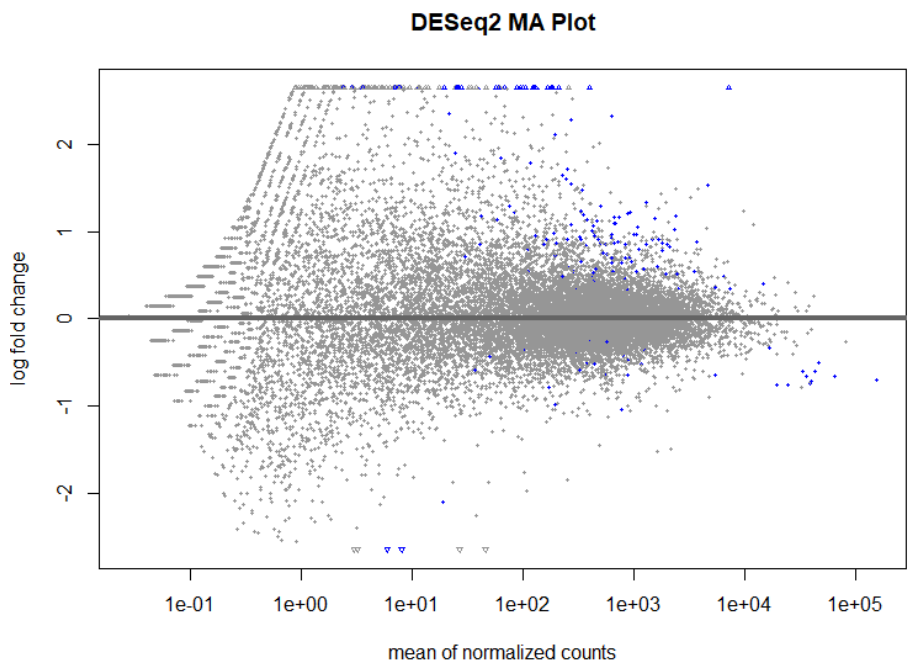
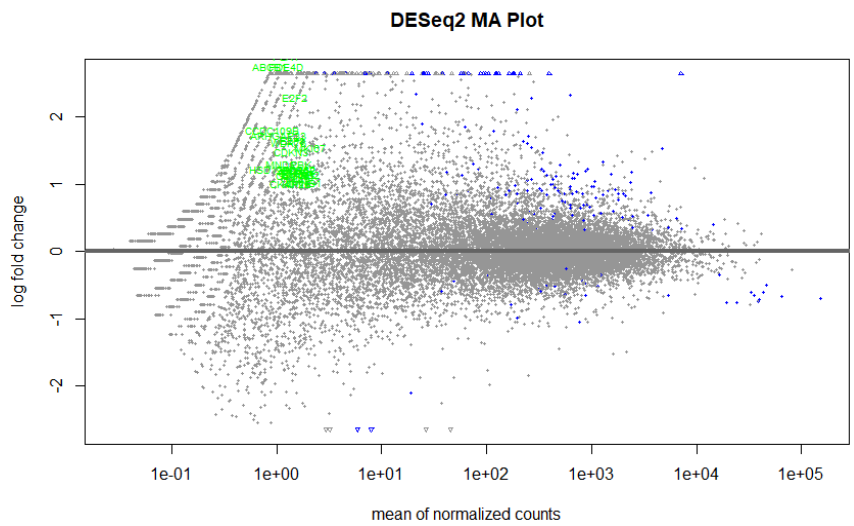
```
# Get results
res <- results(dds)

# View results
head(res)
View(res)



# Plot MA plot
plotMA(res, main="DESeq2 MA Plot")
```

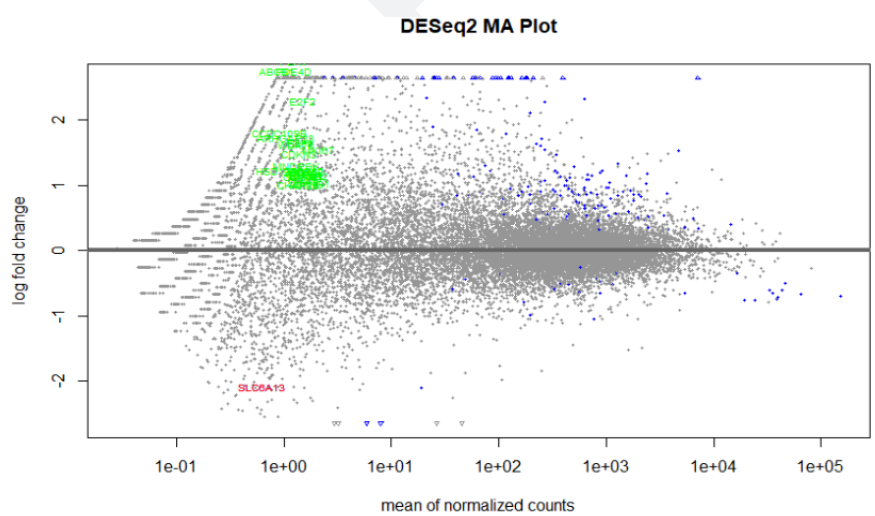



```
# Define upregulated and downregulated genes
upregulated_genes <- rownames(res)[res$log2FoldChange > 1 & res$padj < 0.05]
downregulated_genes <- rownames(res)[res$log2FoldChange < -1 & res$padj < 0.05]

# Add labels for upregulated genes
text(x = log10(res[upregulated_genes, "baseMean"]),
     y = res[upregulated_genes, "log2FoldChange"],
     labels = upregulated_genes,
```

```
    pos = 2,  # Position labels to the right of the points
    col = "green",
    cex = 0.6)  # Adjust label size
```

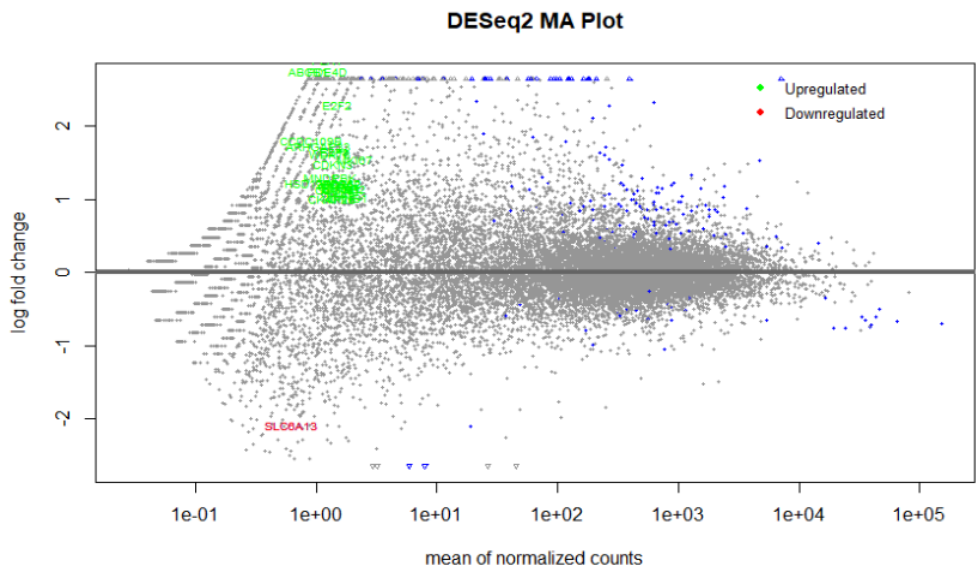



```
# Add labels for downregulated genes
text(x = log10(res[downregulated_genes, "baseMean"]),
    y = res[downregulated_genes, "log2FoldChange"],
    labels = downregulated_genes,
    pos = 2,  # Position labels to the left of the points
    col = "red",
    cex = 0.6)  # Adjust label size
```




```
# Add a legend
legend("topright",                    # Position of the legend
```

```
legend = c("Upregulated", "Downregulated"),  # Legend labels
col = c("green", "red"),          # Colors for the legend
pch = 16,                         # Use solid circles in the legend
cex = 0.8,                        # Adjust legend text size
bty = "n")                        # Remove legend box
```



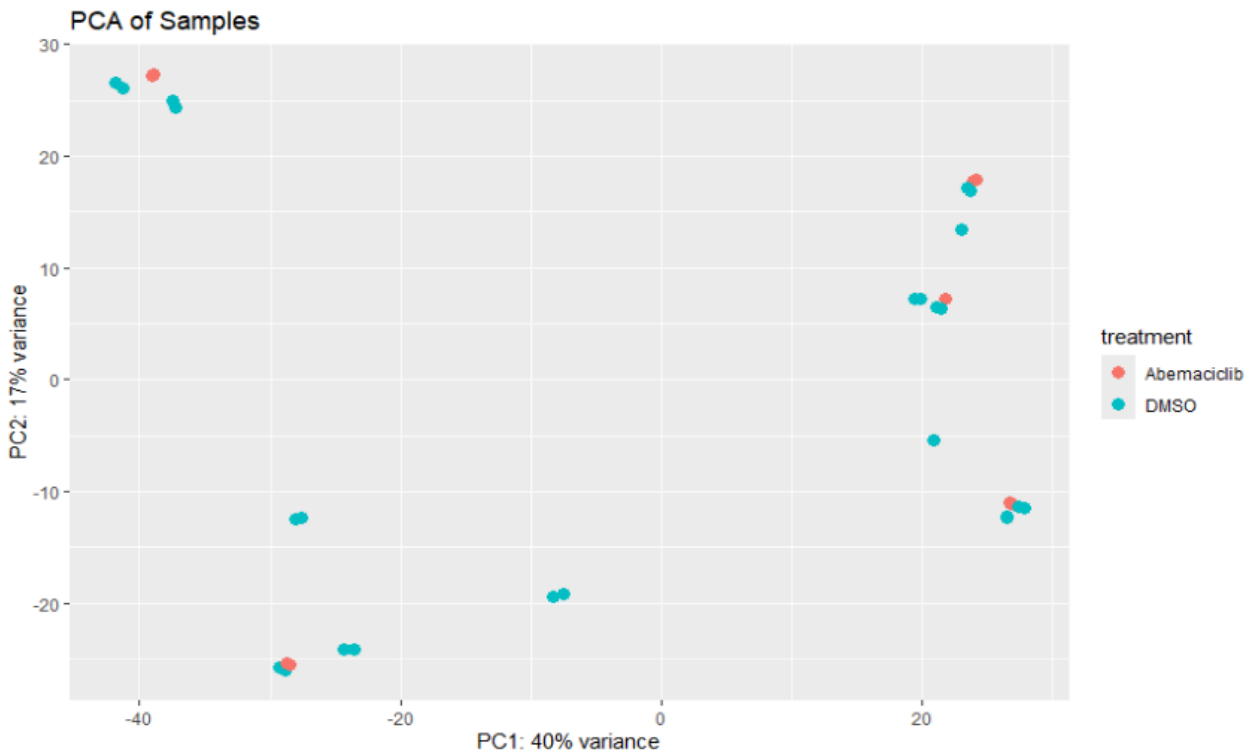
cat("Upregulated Genes:\n", paste(upregulated_genes, collapse = ", "), "\n")

#NA, NA, **AKR1B10**, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, PDE4D, NA, NA, NA, NA, NA, NA, NA, NA, NA, **WDR76**, NA, NA, **WDR62**, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, **KRT81**, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, **ABCB1**, NA, NA, NA, NA, NA, NA, NA, NA, NA, FN1, MAGEA6, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, **CKAP2L**, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, **DMBX1**, NA, NA, NA, **SPC25**, NA, **MAGEC2**, NA, NA, NA, NA, MKI67, NA, NA, NA, **HJURP**, NA, MND1, NA, NA, NA, NA, NA, NA, NA, NA, **SPINK6**, NA, NA, NA, NA, NA, NA, NA, **E2F8,** NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, **RPS4Y1, CDKN3, PLAT**, NA, NA, **HSD17B6,** NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, **RP11-169F17.1,** NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, **CDC45**, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, **FLG,** NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, **ARHGAP33**, NA, NA, NA, **ESPL1**, NA, NA, NA, **KIF2C**, NA, **KIF15**, NA, NA, NA, NA, NA, NA, **LMNB1**, NA, NA, **RUNX1T1**, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, **CCDC109B, CCNA2**, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, **GNG2**, NA, NA, NA, NA, NA, NA, NA, **HMGB2**, NA, NA, NA, NA, NA, NA,

NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, **SP8,** NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, **E2F2**, NA, **CPA4**, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, **DDX3Y**, NA, **ALDH1A2,** NA, **NGFR**, NA, NA, NA, **GTSE1**, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, **PBK,** NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, **MAGEC1**, NA, NA, NA, NA, NA, NA, NA

```
cat("Downregulated Genes:\n", paste(downregulated_genes, collapse = ", "), "\n")
# NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
GOLGA8Q, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, SLC6A13, NA, NA, NA, NA, NA, NA, NA,
NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
IL1A, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA
```

```
# Save the results to a file
write.csv(as.data.frame(res), file="deseq2_results.csv")

# Plot PCA
vsd <- vst(dds, blind = FALSE)  # Variance stabilizing transformation
pcaData <- plotPCA(vsd, intgroup = "treatment", returnData = TRUE)
percentVar <- round(100 * attr(pcaData, "percentVar"))
ggplot(pcaData, aes(PC1, PC2, color = treatment)) +
  geom_point(size = 3) +
  xlab(paste0("PC1: ", percentVar[1], "% variance")) +
  ylab(paste0("PC2: ", percentVar[2], "% variance")) +
  ggtitle("PCA of Samples")
```




```
# Remove rows with NA values in log2FoldChange or padj
res_clean <- res[!is.na(res$log2FoldChange) & !is.na(res$padj),]

# Filter for upregulated genes (log2 fold change > 1 and padj < 0.05)
upregulated_genes <- rownames(res_clean[res_clean$log2FoldChange > 1 & res_clean$padj <
0.05,])

# Filter for downregulated genes (log2 fold change < -1 and padj < 0.05)
downregulated_genes <- rownames(res_clean[res_clean$log2FoldChange < -1 & res_clean$padj
< 0.05,])

# Print the names of the upregulated and downregulated genes
print("Upregulated genes:")
print(upregulated_genes)
# print(upregulated_genes)
#[1] "AKR1B10"      "PDE4D"        "WDR76"        "WDR62"        "KRT81"
#[6] "ABCB1"        "FN1"          "MAGEA6"       "CKAP2L"       "DMBX1"
#[11] "SPC25"        "MAGEC2"       "MKI67"        "HJURP"        "MND1"
#[16] "SPINK6"       "E2F8"         "RPS4Y1"       "CDKN3"        "PLAT"
#[21] "HSD17B6"      "RP11-169F17.1" "CDC45"       "FLG"          "ARHGAP33"
#[26] "ESPL1"        "KIF2C"        "KIF15"        "LMNB1"        "RUNX1T1"
#[31] "CCDC109B"     "CCNA2"        "GNG2"         "HMGB2"        "SP8"
```

```
#[36] "E2F2"        "CPA4"        "DDX3Y"       "ALDH1A2"     "NGFR"
#[41] "GTSE1"       "PBK"         "MAGEC1"

print("Downregulated genes:")
print(downregulated_genes)
#[1] "GOLGA8Q" "SLC6A13" "IL1A"


# Install EnhancedVolcano if not already installed
if (!requireNamespace("BiocManager", quietly = TRUE))
  install.packages("BiocManager")
BiocManager::install("EnhancedVolcano")

# Load EnhancedVolcano library
library(EnhancedVolcano)

# Define upregulated and downregulated genes
upregulated_genes <- rownames(res)[res$log2FoldChange > 1 & res$padj < 0.05]
downregulated_genes <- rownames(res)[res$log2FoldChange < -1 & res$padj < 0.05]

# Create a custom label vector
# Label only upregulated and downregulated genes, leave others blank
custom_labels <- ifelse(rownames(res) %in% upregulated_genes, rownames(res),
                    ifelse(rownames(res) %in% downregulated_genes, rownames(res), ""))

# Create the volcano plot
EnhancedVolcano(res,
            lab = custom_labels,       # Use custom labels for upregulated and downregulated
genes
            x = 'log2FoldChange',      # X-axis: log2 fold change
            y = 'padj',                # Y-axis: adjusted p-value
            xlim = c(-5, 5),           # X-axis limits
            ylim = c(0, 10),           # Y-axis limits
            pCutoff = 0.05,            # p-value cutoff
            FCcutoff = 1,              # log2 fold change cutoff
            title = "Volcano Plot",    # Title of the plot
            subtitle = NULL,           # Remove subtitle
            col = c('grey', 'red', 'green'),  # All points are grey
            legendLabels = c('NS', 'Upregulated', 'Downregulated'),  # Legend labels
            labSize = 3,               # Size of the labels
```

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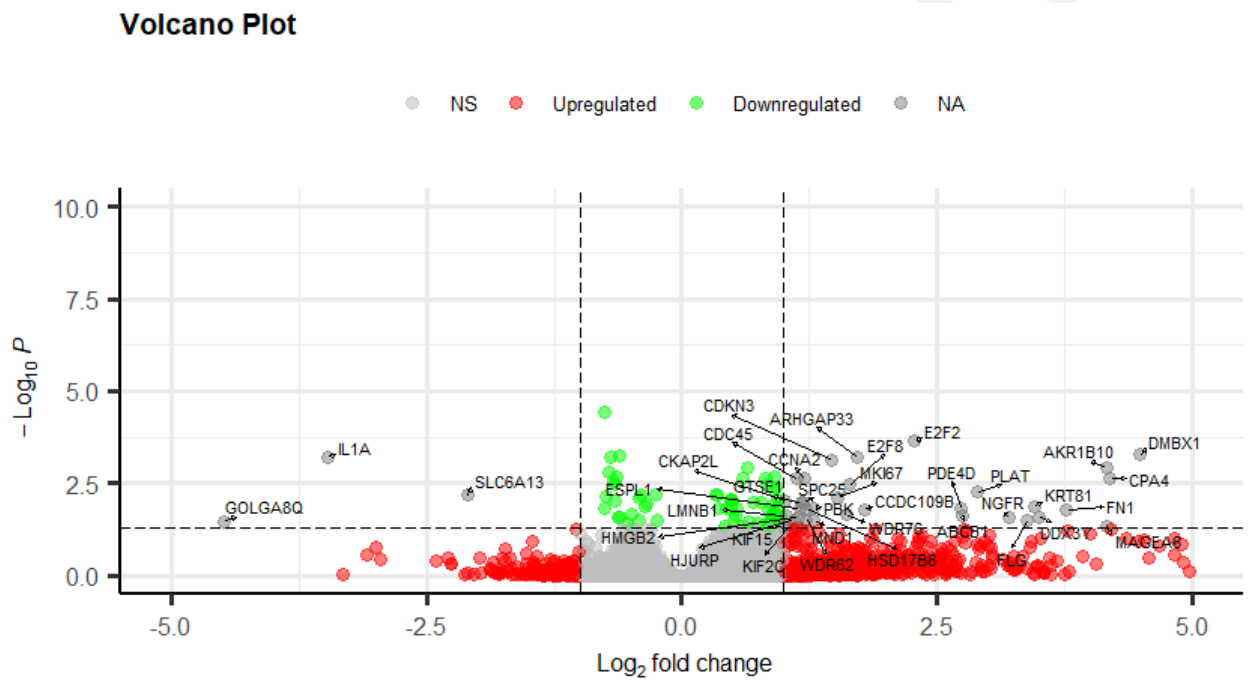
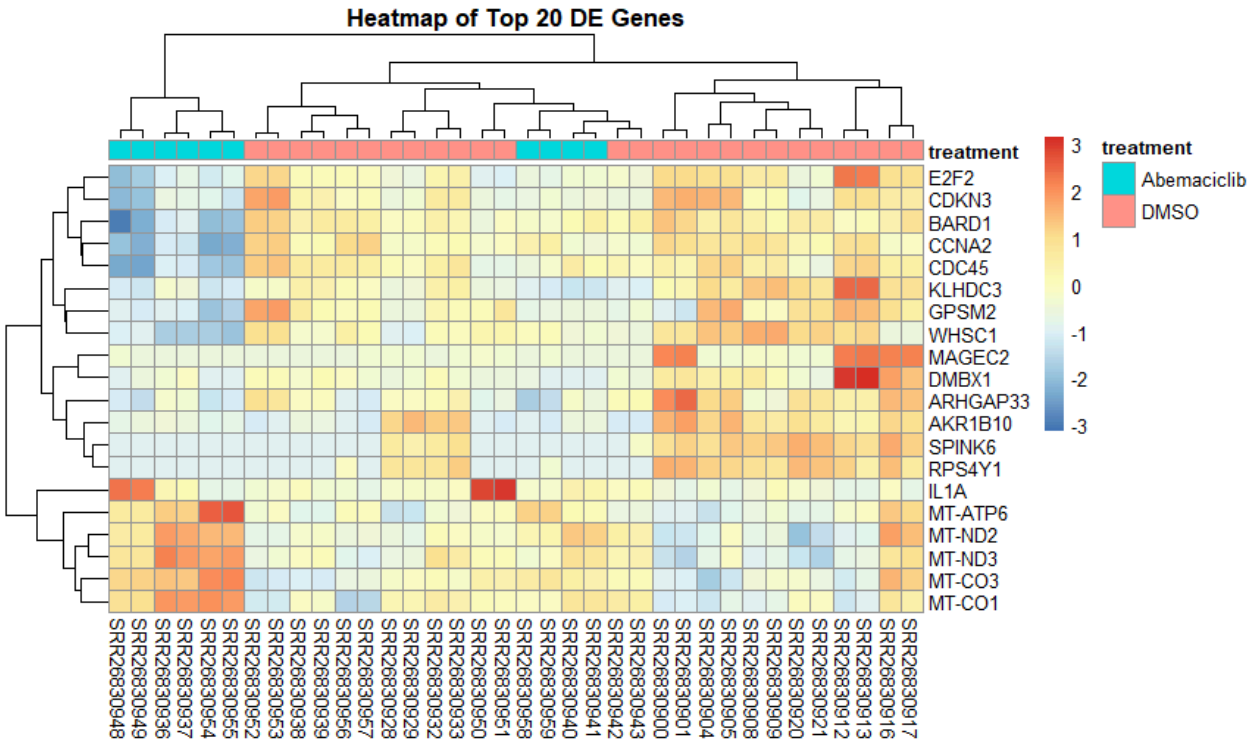
```
        pointSize = 3,            # Size of the points (balls)
        shape = 19,               # Shape of the points (19 = solid circle)
        legendPosition = 'top',   # Position of the legend (top of the figure)
        legendLabSize = 10,       # Size of legend labels
        legendIconSize = 3,       # Size of icons in the legend
        drawConnectors = TRUE,    # Draw connectors between labels and points
        widthConnectors = 0.5,    # Width of the connector lines
        colConnectors = 'black',  # Color of the connector lines
        axisLabSize = 12,         # Size of x and y axis labels
        titleLabSize = 14,        # Size of the title
        max.overlaps = Inf        # Allow all labels to be displayed
)
```




```
## Select top 20 differentially expressed genes by adjusted p-value
topGenes <- head(order(res$padj), 20)

# Plot heatmap
library(pheatmap)
pheatmap(assay(vsd)[topGenes,],
        cluster_rows = TRUE,
        cluster_cols = TRUE,
        show_rownames = TRUE,
        show_colnames = TRUE,
```

```
annotation_col = meta_data_edit[, "treatment", drop = FALSE],
scale = "row",
main = "Heatmap of Top 20 DE Genes")
```



###pathway enrichment analysis#########

```
  # Install and load required packages
  if (!requireNamespace("BiocManager", quietly = TRUE, force = TRUE))
    install.packages("BiocManager")
BiocManager::install("clusterProfiler")
BiocManager::install("org.Hs.eg.db")

library(clusterProfiler)
library(org.Hs.eg.db)

# Define gene lists
upregulated_genes <- c("AKR1B10", "PDE4D", "WDR76", "WDR62", "KRT81", "ABCB1",
"FN1", "MAGEA6", "CKAP2L", "DMBX1",
```

```
                "SPC25", "MAGEC2", "MKI67", "HJURP", "MND1", "SPINK6", "E2F8",
"RPS4Y1", "CDKN3", "PLAT",
                "HSD17B6", "RP11-169F17.1", "CDC45", "FLG", "ARHGAP33", "ESPL1",
"KIF2C", "KIF15", "LMNB1",
                "RUNX1T1", "CCDC109B", "CCNA2", "GNG2", "HMGB2", "SP8", "E2F2",
"CPA4", "DDX3Y", "ALDH1A2",
                "NGFR", "GTSE1", "PBK", "MAGEC1")

downregulated_genes <- c("GOLGA8Q", "SLC6A13", "IL1A")

# Convert gene symbols to Entrez IDs
upregulated_entrez <- bitr(upregulated_genes, fromType = "SYMBOL", toType =
"ENTREZID", OrgDb = org.Hs.eg.db)
downregulated_entrez <- bitr(downregulated_genes, fromType = "SYMBOL", toType =
"ENTREZID", OrgDb = org.Hs.eg.db)

# Extract Entrez IDs
upregulated_entrez_ids <- upregulated_entrez$ENTREZID
downregulated_entrez_ids <- downregulated_entrez$ENTREZID

# View the results
print("Upregulated Entrez IDs:")
print(upregulated_entrez_ids)
#[1] "57016"  "5144"   "79968"  "284403" "3887"   "5243"   "2335"   "4105"   "150468"
#[10] "127343" "57405"  "51438"  "4288"   "55355"  "84057"  "404203" "79733"  "6192"
#[19] "1033"   "5327"   "8630"   "8318"   "2312"   "115703" "9700"   "11004"  "56992"
#[28] "4001"   "862"    "890"    "54331"  "3148"   "221833" "1870"   "51200"  "8653"
#[37] "8854"   "4804"   "51512"  "55872"  "9947"

print("Downregulated Entrez IDs:")
print(downregulated_entrez_ids)
#[1] "727909" "6540"   "3552"
```

Perform GO enrichment for upregulated genes

```
go_up <- enrichGO(gene = upregulated_entrez_ids,
                  OrgDb = org.Hs.eg.db,
                  keyType = "ENTREZID",
                  ont = "BP",  # Biological Process
                  pAdjustMethod = "BH",  # Benjamini-Hochberg correction
                  pvalueCutoff = 0.05,
                  qvalueCutoff = 0.05)

# Perform GO enrichment for downregulated genes
go_down <- enrichGO(gene = downregulated_entrez_ids,
                    OrgDb = org.Hs.eg.db,
                    keyType = "ENTREZID",
                    ont = "BP",  # Biological Process
                    pAdjustMethod = "BH",  # Benjamini-Hochberg correction
                    pvalueCutoff = 0.05,
                    qvalueCutoff = 0.05)

# Extract key columns from the GO enrichment results
go_results <- as.data.frame(go_up)[, c("ID", "Description", "p.adjust", "geneID")]

# Rename columns for clarity
colnames(go_results) <- c("GO Term ID", "GO Term Description", "Adjusted p-value", "Genes")

# Sort by adjusted p-value (most significant terms first)
go_results <- go_results[order(go_results$`Adjusted p-value`), ]

# Display the top 10 results
print(head(go_results, 10))

# Convert Entrez IDs to gene symbols
gene_symbols <- bitr(unlist(strsplit(go_results$Genes, "/")), fromType = "ENTREZID", toType
= "SYMBOL", OrgDb = org.Hs.eg.db)

# Replace Entrez IDs with gene symbols in the results
go_results$Genes <- sapply(strsplit(go_results$Genes, "/"), function(ids) {
  symbols <- gene_symbols$SYMBOL[match(ids, gene_symbols$ENTREZID)]
  paste(symbols, collapse = ", ")
})

# Display the cleaned results
```

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```
print(head(go_results, 10))


# Save the cleaned results to a CSV file
write.csv(go_results, file = "UPREGULATED_go_enrichment_cleaned_results.csv", row.names
= FALSE)
```

```
  # Extract key columns from the GO enrichment results
  go_down_results <- as.data.frame(go_down)[, c("ID", "Description", "p.adjust", "geneID")]

# Rename columns for clarity
colnames(go_down_results) <- c("GO Term ID", "GO Term Description", "Adjusted p-value",
"Genes")

# Sort by adjusted p-value (most significant terms first)
go_down_results <- go_down_results[order(go_down_results$`Adjusted p-value`), ]

# Convert Entrez IDs to gene symbols
gene_symbols_down <- bitr(unlist(strsplit(go_down_results$Genes, "/")), fromType =
"ENTREZID", toType = "SYMBOL", OrgDb = org.Hs.eg.db)

# Replace Entrez IDs with gene symbols in the results
go_down_results$Genes <- sapply(strsplit(go_down_results$Genes, "/"), function(ids) {
  symbols <- gene_symbols_down$SYMBOL[match(ids, gene_symbols_down$ENTREZID)]
  paste(symbols, collapse = ", ")
})

# Display the cleaned results
print(head(go_down_results, 10))

# Save the cleaned results to a CSV file
write.csv(go_down_results, file = "DOWN_REGULATED_enrichment_cleaned_results.csv",
row.names = FALSE)
```

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